EE4213: HUMAN-COMPUTER INTERACTION

Effective Term
Semester A 2025/26

Part I Course Overview

Course Title
Human-Computer Interaction

Subject Code
EE - Electrical Engineering
Course Number
4213

Academic Unit
Electrical Engineering (EE)

College/School
College of Engineering (EG)

Course Duration
One Semester

Credit Units
3

Level
B1, B2, B3, B4 - Bachelor's Degree

Medium of Instruction
English

Medium of Assessment
English

Prerequisites
CS2311 Computer Programming
or
equivalent

Precursors
Nil

Equivalent Courses
Nil

Exclusive Courses
Nil

Part II Course Details

Abstract

This course explores Human-Computer Interaction (HCI) in the context of today's AI-driven digital transformation. Students will learn foundational HCI principles while gaining hands-on experience with cutting-edge tools for design, analytics, and automation. The curriculum combines theoretical frameworks with practical application through real-world product development cycles aligned with industry standards. Students will develop essential skills in user research, prototyping, usability evaluation, and Agile methodologies. As AI reshapes the technology landscape, emphasis is placed on developing higher-level capabilities such as strategic decision-making, user experience design, and cross-disciplinary collaboration. Through individual and group projects, students will learn to create impactful, human-centered digital solutions that leverage AI in product development while addressing real user needs in a rapidly evolving technological environment.

Course Intended Learning Outcomes (CILOs)

	CILOs	Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Analyze key principles and theories of human-computer interaction and their application in interaction design processes.		x	x	
2	Evaluate the usability of user interfaces using established HCI criteria and guidelines		x	x	
3	Apply human-computer interaction principles to critically assess and improve the design of everyday digital products and interfaces.		x	x	
4	Design and prototype user-centered interfaces that address specific usability challenges and user needs.				x
5	Conduct basic user research and usability testing to inform and validate design decisions.		x	x	x

A1: Attitude
Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability
Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments
Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

	LTAs	Brief Description	CILO No.	Hours/week (if applicable)
1	Lectures	Students engage with core HCI concepts through instructor-led discussions, multimedia presentations, and in-class exercises on interface design principles and usability evaluation methods. Key concepts are worked out based on problems, in-class Q&A, work-along examplesrated	1, 2, 3, 4, 5	3 hrs/week
2	Workshops	Students participate in hands-on sessions that combine technical tutorials and design activities. These workshops focus on implementing specific HCI techniques, working on both individual and group design challenges, and engaging in peer feedback and collaborative problem-solving.	1, 2, 3, 4, 5	

Assessment Tasks / Activities (ATs)

	ATs	CILO No.	Weighting (%)	Remarks ("-" for nil entry)	Allow Use of GenAI?
1	Tests (min.: 1)	1, 2, 3, 4, 5	10	-	No
2	Projects	1, 2, 3, 4, 5	60	-	Yes

Continuous Assessment (%)

70

Examination (%)

30

Examination Duration (Hours)

2

Minimum Continuous Assessment Passing Requirement (%)

30

Minimum Examination Passing Requirement (%)

30

Assessment Rubrics (AR)

Assessment Task

Examination

Criterion
Achievements in CILOs

Excellent (A+, A, A-)
High

Good (B+, B, B-)
Significant

Fair (C+, C, C-)
Moderate

Marginal (D)
Basic

Failure (F)
Not even reaching marginal levels

Assessment Task
Coursework

Criterion
Achievements in CILOs

Excellent (A+, A, A-)
High

Good (B+, B, B-)
Significant

Fair (C+, C, C-)
Moderate

Marginal (D)
Basic

Failure (F)
Not even reaching marginal levels

Part III Other Information

Keyword Syllabus

Foundations of Human Computer Interaction
Overview of Human Computer Interaction; Psychology and cognitive models; Human factors; Design thinking framework; Sensory systems and perception; Attention and memory in interface design

User-Centered Design Process
User research methodologies; Data analytics for user insights; Persona development; Problem identification and framing; Prototyping (low and high-fidelity); Usability design principles; Generative AI for interface design; Evaluation methods and optimization

Agile Development and Implementation for HCI

Agile methodologies with focus on Scrum; User stories and acceptance criteria; Sprint planning and backlog management; AI tools for Agile processes; Collaborative product development; Iterative design and continuous improvement

Advanced Interaction Modalities and Applications

Gesture recognition and tracking; Voice and natural language interfaces; Mobile and responsive design; Accessibility and inclusive design; AI-enhanced interfaces; Evaluation metrics and usability testing; Industry applications and case studies

Reading List

Compulsory Readings

	Title
1	Nil

Additional Readings

	Title
1	D.Norman, The Design of Everyday Things, Basic Books (2013)
2	J.Preece, Y.Roger and H.Sharp, Interaction Design: Beyond Human-Computer Interaction, 5th Edition, Wiley (2019)
3	Dix, Finaly, Abowd and Beale, Human-Computer Interaction (3rd Edition), Pearson, 2003.